

Question

At 273K, a mass m of NaCl solid (whose specific heat can be neglected) is sprinkled into a mixture containing 100 g of ice and 100 g of water. Assuming the formed solution is an ideal mixture, neglecting the thermal effect of NaCl dissolution and heat exchange between the mixture and the surroundings, calculate the required mass m of NaCl and the mass of ice at this point to ensure the final temperature of the ice-water mixture is 268K.

Data that may be used for this problem:

The specific heat capacity of water is $4.18 \text{ J}/(\text{g} \cdot \text{K})$,

The specific heat capacity of ice is $2.10 \text{ J}/(\text{g} \cdot \text{K})$,

The enthalpy of fusion of ice at 298K is 6008 J/mol .

Hint: When the amount of NaCl added is large, using approximate formulas may lead to errors.

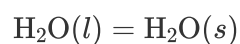
- A. The mass of NaCl required is 7.78 g, and the mass of ice is 88.5 g.
- B. The required mass of NaCl is 17.3 g, and the mass of ice is 90.3 g.
- C. The required mass of NaCl is 15.5 g, and the mass of ice is 88.5 g.
- D. The required mass of NaCl is 9.78 g, and the mass of ice is 68.5 g.
- E. The required mass of NaCl is 8.61 g, and the mass of ice is 90.3 g.
- F. None of the above results are correct (with an error greater than 5%).

Answer

Correct Answer: **A**

Detailed Explanation

In this problem, due to the large mass of NaCl, directly applying the freezing point depression law would cause errors!



$$\Delta_r C_m = -37.47 \text{ J/mol}\cdot\text{K}$$

At 273K , the equilibrium constant of the above reaction is 1. Using Kirchhoff's law, the equilibrium constant at 268K can be calculated as follows:

$$K_2 = 1.043 = [n(\text{H}_2\text{O}, l) + 2n(\text{NaCl})]/n(\text{H}_2\text{O}, l)$$

CHECKPOINT

2 PTS

$$K_2 = 1.043 = [n(\text{H}_2\text{O}, l) + 2n(\text{NaCl})]/n(\text{H}_2\text{O}, l)$$

Additionally, since $\Delta H = 0$ throughout the process, we have:

$$0 = 5071.2[n(\text{H}_2\text{O}, l) - n_0(\text{H}_2\text{O}, l)] + C \times (268\text{K} - 273\text{K})$$

CHECKPOINT

2 PTS

$$0 = 5071.2[n(\text{H}_2\text{O}, l) - n_0(\text{H}_2\text{O}, l)] + C \times (268\text{K} - 273\text{K})$$

Solving the two equations simultaneously yields: $n(\text{H}_2\text{O}, s) = 4.91 \text{ mol}$, , $n(\text{NaCl}) = 0.133\text{mol}$

CHECKPOINT

2 PTS

$$n(\text{H}_2\text{O}, s) = 4.91 \text{ mol}, , n(\text{NaCl}) = 0.133\text{mol}$$

The required mass of NaCl is 7.78g , and the mass of ice is 88.5g .